

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, C++, JavaScript, Java, SQL

Tools and Frameworks: REST(Django, FastAPI), React, Next.js, PostgreSQL, CMake, OpenGL, Unreal Engine, AWS, Redis, Kafka, Docker, Kubernetes, GitHub Actions(CI/CD), Git, Linux, PyTorch, LangChain, Pinecone

Areas of Interest and Coursework: Data Structures & Algorithms, Computer Graphics, AI & LLMs

INDUSTRY EXPERIENCE

Data AI Intern | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Developed backend services for an AI-native GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from days of manual work to <10 minutes.
- Diagnosed an **out-of-memory** failure in the fallback path, tracing it to freshly pulled 4–5 GB model weights lingering in the OS page cache, and fixed it with targeted **posix_fadvise** eviction, **eliminating model-load failures**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an AI-powered PRD review platform in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from 8 to 2 hours.

Full Stack Developer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to Kafka workers for order confirmation, customer notification, and crew allocation, taking \$600K+ in online bookings across ~1,100 orders.
- Designed and deployed the AWS architecture that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10× seasonal peaks and load-tested to ~50 quote requests/sec at <120 ms p99 with zero-downtime rolling deploys.

ACADEMIC EXPERIENCE / PROJECTS

Graduate Assistant | UMBC - Data Structures & Software Engineering

Jan 2025 – June 2026

- Designed a **C++ assignment** on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.
- Developed **200+** unit tests for the DSA projects on Linked Lists, Self Balancing Trees, Heaps, and Hash Tables automating logic checking and reducing grading time by over **80%**.
- Helped **50+** students design and debug **C++** projects implementing core data structures, and graded their homework.
- Held office hours to clarify technical concepts and translate requirements into actionable implementation steps.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

Rendering Pipelines from Scratch | Ray tracer, rasterizer, and real-time OpenGL renderer in C++

- Built a **C++ ray tracer** that parses Polyray scenes and shades them with Blinn-Phong direct lighting, recursive reflection, and Schlick-Fresnel refraction, then accelerated it with a median-split **BVH** over primitive centroids and an atomic row-chunk thread pool, cutting a 512×512 render from **106 s to 0.14 s (~759×)**.
- Implemented **DXT1 texture compression** end to end in an OpenGL scene renderer, fitting each 4×4 texel block's palette by principal-axis covariance, packing 2-bit selectors into the 64-bit block layout, and compressing every CPU-generated mip level independently, shrinking color textures **6× in GPU memory** behind a runtime A/B toggle.

Bloodborne TAG | Text Adventure Game with natural language command parsing

- Architected a data-driven **game engine** that loads a **41-room** world and **60+** items/NPCs from JSON into in-memory session state, gating progression on generic key-item fields so new content ships with **zero** engine code changes.
- Engineered a **command parser** using few-shot Mistral-7B for slot extraction and embedding-similarity-based intent matching, mapping free-form playtest commands to 14 deterministic actions at **96%** intent accuracy.